PADHAKU
Mathematics · Class 10 Mathematics

CHAPTER 6

Triangles

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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❓ In-text questions (Why? / How?)

6.2 **Can a circle and a square be similar? Can a triangle and a square be similar? (Why?)**

✔ ANSWER

No, a circle and a square are not similar, and a triangle and a square are not similar.

Similar figures must have the **same shape**. A circle has no sides or corners, a square has 4 sides and a triangle has 3 sides, so their shapes are different.

Fig. 6.2 **What can you say about the two quadrilaterals ABCD and PQRS (Fig. 6.2)? Are they similar?**

✔ ANSWER

They look similar, but we cannot be sure just by looking.

To decide, we must measure and check both conditions: (i) corresponding angles are equal and (ii) corresponding sides are in the same ratio.

Activity 2 **In Δ ABC, DE || BC and AD/DB = AE/EC. Is it a coincidence?**

✔ ANSWER

No. It is because of the **Basic Proportionality Theorem** (Theorem 6.1): a line drawn parallel to one side of a triangle divides the other two sides in the same ratio.

Theorem 6.2

In the proof of Theorem 6.2: (1) Why AD/DB = AE′/E′C? (2) Why AE/EC = AE′/E′C? (3) Why must E and E′ coincide?

✔ ANSWER

(1) DE′ || BC, so by Theorem 6.1 (BPT), AD/DB = AE′/E′C.

(2) It is given that AD/DB = AE/EC. Both ratios equal AD/DB, so AE/EC = AE′/E′C.

(3) Adding 1 to both sides: (AE + EC)/EC = (AE′ + E′C)/E′C, i.e. AC/EC = AC/E′C.

So EC = E′C. Both E and E′ lie on AC at the same distance from C, so **E and E′ are the same point**. Hence DE || BC.

Theorem 6.3

In the proof of Theorem 6.3 (AAA): Why Δ ABC ≅ Δ DPQ? How PQ || EF? Why DP/PE = DQ/QF and AB/DE = AC/DF?

✔ ANSWER

Δ ABC ≅ Δ DPQ by **SAS congruence**: DP = AB, DQ = AC (by construction) and ∠A = ∠D (given).

So ∠B = ∠P (CPCT). Also ∠B = ∠E (given), so ∠P = ∠E. These are corresponding angles made by PQ and EF with DE, so **PQ || EF**.

PQ || EF in Δ DEF, so by BPT, DP/PE = DQ/QF.

Adding 1 to both sides: DE/DP = DF/DQ, i.e. DP/DE = DQ/DF. As DP = AB and DQ = AC, **AB/DE = AC/DF**.

Theorem 6.4 **In the proof of Theorem 6.4 (SSS): How DP/PE = DQ/QF and PQ || EF? Why DP/DE = DQ/DF = BC/EF? Why BC = PQ? Why Δ ABC ≅ Δ DPQ? How are the angles equal?**

✔ ANSWER

Given AB/DE = AC/DF and DP = AB, DQ = AC, so DP/DE = DQ/DF. Then PE/DP = QF/DQ (subtracting 1 from reciprocals), i.e. DP/PE = DQ/QF. So by Theorem 6.2, **PQ || EF**.

Hence ∠P = ∠E and ∠Q = ∠F (corresponding angles), so Δ DPQ ~ Δ DEF (AA) and DP/DE = DQ/DF = PQ/EF.

Also AB/DE = BC/EF (given) and AB = DP, so DP/DE = BC/EF. Therefore PQ/EF = BC/EF, giving **BC = PQ**.

Now AB = DP, AC = DQ, BC = PQ, so Δ ABC ≅ Δ DPQ (**SSS congruence**).

So ∠A = ∠D, ∠B = ∠P = ∠E and ∠C = ∠Q = ∠F (CPCT and corresponding angles).

Theorem 6.5 **In the proof of Theorem 6.5 (SAS): How PQ || EF and Δ ABC ≅ Δ DPQ? Why Δ ABC ~ Δ DEF?**

✔ ANSWER

DP = AB, DQ = AC, so from AB/DE = AC/DF we get DP/DE = DQ/DF, i.e. DP/PE = DQ/QF. By Theorem 6.2, **PQ || EF**.

Δ ABC ≅ Δ DPQ by **SAS congruence** (DP = AB, DQ = AC, ∠A = ∠D).

So ∠B = ∠P and ∠C = ∠Q (CPCT); and ∠P = ∠E, ∠Q = ∠F (corresponding angles, PQ || EF).

Thus ∠A = ∠D, ∠B = ∠E, ∠C = ∠F, so **Δ ABC ~ Δ DEF** by the AAA similarity criterion.

Exercise 6.1

1 **Fill in the blanks using the correct word given in brackets: (i) All circles are ____. (congruent, similar) (ii) All squares are ____. (similar, congruent) (iii) All ____ triangles are similar. (isosceles, equilateral) (iv) Two polygons of the same number of sides are similar, if (a) their corresponding angles are ____ and (b) their corresponding sides are ____. (equal, proportional)**

✔ ANSWER

(i) All circles are **similar**.

(ii) All squares are **similar**.

(iii) All **equilateral** triangles are similar.

(iv) (a) corresponding angles are **equal** and (b) corresponding sides are **proportional**.

2 **Give two different examples of pair of (i) similar figures (ii) non-similar figures.**

✔ ANSWER

(i) Similar figures: any two circles (e.g. of radii 2 cm and 5 cm); any two squares (e.g. of sides 3 cm and 7 cm). (Two equilateral triangles also work.)

(ii) Non-similar figures: a circle and a square; a triangle and a rectangle. (A square and a rhombus, or a square and a rectangle, are also non-similar.)

3 **State whether the following quadrilaterals are similar or not (Fig. 6.8: PQRS with all sides 1.5 cm but angles not 90°; ABCD a square of side 3 cm).**

✔ ANSWER

Corresponding sides: PQ/AB = QR/BC = RS/CD = SP/DA = 1.5/3 = ½ (same ratio).

But the angles of square ABCD are 90° each, while the angles of PQRS are not 90° (it is a rhombus, not a square).

Corresponding angles are not equal, so the quadrilaterals are **not similar**.

Exercise 6.2

1 **In Fig. 6.17, (i) and (ii), DE || BC. Find EC in (i) and AD in (ii). (i) AD = 1.5 cm, DB = 3 cm, AE = 1 cm. (ii) DB = 7.2 cm, AE = 1.8 cm, EC = 5.4 cm.**

✔ ANSWER

DE || BC, so by the Basic Proportionality Theorem (BPT), AD/DB = AE/EC.

(i) 1.5/3 = 1/EC ⇒ EC = 3/1.5 = **2 cm**.

(ii) AD/7.2 = 1.8/5.4 ⇒ AD = (7.2 × 1.8)/5.4 = 7.2/3 = **2.4 cm**.

2 **E and F are points on the sides PQ and PR respectively of a Δ PQR. For each of the following cases, state whether EF || QR: (i) PE = 3.9 cm, EQ = 3 cm, PF = 3.6 cm and FR = 2.4 cm (ii) PE = 4 cm, QE = 4.5 cm, PF = 8 cm and RF = 9 cm (iii) PQ = 1.28 cm, PR = 2.56 cm, PE = 0.18 cm and PF = 0.36 cm**

✔ ANSWER

By the converse of BPT (Theorem 6.2), EF || QR if PE/EQ = PF/FR.

(i) PE/EQ = 3.9/3 = 1.3; PF/FR = 3.6/2.4 = 1.5. Not equal ⇒ **EF is not parallel to QR**.

(ii) PE/QE = 4/4.5 = 8/9; PF/RF = 8/9. Equal ⇒ **EF || QR**.

(iii) EQ = PQ − PE = 1.28 − 0.18 = 1.10 cm; FR = PR − PF = 2.56 − 0.36 = 2.20 cm.

PE/EQ = 0.18/1.10 = 9/55; PF/FR = 0.36/2.20 = 9/55. Equal ⇒ **EF || QR**.

3 **In Fig. 6.18, if LM || CB and LN || CD, prove that AM/AB = AN/AD.**

✔ ANSWER

1. In Δ ABC, LM || CB. By BPT (in the form of Example 1), AM/AB = AL/AC. ... (1)
2. In Δ ACD, LN || CD. By BPT, AN/AD = AL/AC. ... (2)
3. From (1) and (2), **AM/AB = AN/AD**. Proved.

4 **In Fig. 6.19, DE || AC and DF || AE. Prove that BF/FE = BE/EC.**

✔ ANSWER

1. In Δ ABC, DE || AC. By BPT, BD/DA = BE/EC. ... (1)
2. In Δ ABE, DF || AE. By BPT, BD/DA = BF/FE. ... (2)
3. From (1) and (2), **BF/FE = BE/EC**. Proved.

5 **In Fig. 6.20, DE || OQ and DF || OR. Show that EF || QR.**

✔ ANSWER

1. In Δ POQ, DE || OQ. By BPT, PE/EQ = PD/DO. ... (1)

2. In Δ POR, DF || OR. By BPT, PF/FR = PD/DO. ... (2)

3. From (1) and (2), PE/EQ = PF/FR.

4. In Δ PQR, the line EF divides sides PQ and PR in the same ratio, so by the converse of BPT (Theorem 6.2), **EF || QR**.

6 **In Fig. 6.21, A, B and C are points on OP, OQ and OR respectively such that AB || PQ and AC || PR. Show that BC || QR.**

✔ ANSWER

1. In Δ OPQ, AB || PQ. By BPT, OA/AP = OB/BQ. ... (1)

2. In Δ OPR, AC || PR. By BPT, OA/AP = OC/CR. ... (2)

3. From (1) and (2), OB/BQ = OC/CR.

4. In Δ OQR, BC divides OQ and OR in the same ratio, so by Theorem 6.2, **BC || QR**.

7 **Using Theorem 6.1, prove that a line drawn through the mid-point of one side of a triangle parallel to another side bisects the third side.**

✔ ANSWER

Given: Δ ABC, D is the mid-point of AB (AD = DB), and DE || BC meets AC at E. To prove: AE = EC.

1. DE || BC, so by BPT (Theorem 6.1), AD/DB = AE/EC.

2. AD = DB, so AD/DB = 1. Hence AE/EC = 1.

3. So **AE = EC**, i.e. DE bisects AC. Proved.

8 **Using Theorem 6.2, prove that the line joining the mid-points of any two sides of a triangle is parallel to the third side.**

✔ ANSWER

Given: Δ ABC, D and E are mid-points of AB and AC. To prove: DE || BC.

1. AD = DB ⇒ AD/DB = 1; AE = EC ⇒ AE/EC = 1.

2. So AD/DB = AE/EC: DE divides AB and AC in the same ratio.

3. By the converse of BPT (Theorem 6.2), **DE || BC**. Proved.

9 **ABCD is a trapezium in which AB || DC and its diagonals intersect each other at the point O. Show that AO/BO = CO/DO.**

✔ ANSWER

Construction: through O draw a line EO || DC, meeting AD at E. Then EO || AB also (AB || DC).

1. In Δ ADC, EO || DC. By BPT, AE/ED = AO/OC. ... (1)

2. In Δ DAB, EO || AB. By BPT, DE/EA = DO/OB, i.e. AE/ED = BO/OD. ... (2)

3. From (1) and (2), AO/OC = BO/OD.

4. Rearranging, **AO/BO = CO/DO**. Proved.

10 **The diagonals of a quadrilateral ABCD intersect each other at the point O such that AO/BO = CO/DO. Show that ABCD is a trapezium.**

✔ ANSWER

Construction: through O draw EO || AB, meeting AD at E.

1. In Δ DAB, EO || AB. By BPT, DE/EA = DO/OB. ... (1)

2. Given AO/BO = CO/DO ⇒ DO/OB = CO/AO. ... (2)

3. From (1) and (2), DE/EA = CO/AO.

4. In Δ ADC, EO divides AD and AC in the same ratio (DE/EA = CO/OA), so by Theorem 6.2, EO || DC.

5. EO || AB and EO || DC ⇒ AB || DC. A quadrilateral with one pair of opposite sides parallel is a trapezium. So **ABCD is a trapezium**. Proved.

📝 Exercise 6.3

1 **State which pairs of triangles in Fig. 6.34 are similar. Write the similarity criterion used and the pairs of similar triangles in symbolic form. (i) Δ ABC: ∠A = 60°, ∠B = 80°, ∠C = 40°; Δ PQR: ∠P = 60°, ∠Q = 80°, ∠R = 40° (ii) Δ**

ABC: AB = 2, BC = 2.5, CA = 3; Δ PQR: QR = 4, RP = 5, PQ = 6 (iii) Δ LMP: LM = 2.7, MP = 2, LP = 3; Δ DEF: DE = 4, EF = 5, DF = 6 (iv) Δ MNL: MN = 2.5, ML = 5, ∠M = 70°; Δ QPR: PQ = 5, QR = 10, ∠Q = 70° (v) Δ ABC: AB = 2.5, BC = 3, ∠A = 80°; Δ DEF: DF = 5, EF = 6, ∠F = 80° (vi) Δ DEF: ∠D = 70°, ∠E = 80°; Δ PQR: ∠Q = 80°, ∠R = 30°

✔ ANSWER

(i) ∠A = ∠P = 60°, ∠B = ∠Q = 80°, ∠C = ∠R = 40°. **Yes, AAA criterion; Δ ABC ~ Δ PQR.**

(ii) AB/QR = 2/4 = ½, BC/RP = 2.5/5 = ½, CA/PQ = 3/6 = ½. All sides in the same ratio. **Yes, SSS criterion; Δ ABC ~ Δ QRP.**

(iii) MP/DE = 2/4 = ½, LP/DF = 3/6 = ½, but LM/EF = 2.7/5 = 0.54 ≠ ½. **No, not similar.**

(iv) MN/QP = 2.5/5 = ½, ML/QR = 5/10 = ½ and the included angles ∠M = ∠Q = 70°. **Yes, SAS criterion; Δ MNL ~ Δ QPR.**

(v) AB/DF = 2.5/5 = ½, BC/EF = 3/6 = ½, but the 80° angle is at A in Δ ABC (not between AB and BC) while it is at F in Δ DEF (between DF and EF). The equal angle is not the included angle. **No, not similar.**

(vi) In Δ DEF, ∠F = 180° − 70° − 80° = 30°. In Δ PQR, ∠P = 180° − 80° − 30° = 70°. So ∠D = ∠P = 70°, ∠E = ∠Q = 80°, ∠F = ∠R = 30°. **Yes, AA criterion; Δ DEF ~ Δ PQR.**

2 **In Fig. 6.35, Δ ODC ~ Δ OBA, ∠BOC = 125° and ∠CDO = 70°. Find ∠DOC, ∠DCO and ∠OAB.**

✔ ANSWER

DOB is a straight line, so ∠DOC + ∠BOC = 180° ⇒ ∠DOC = 180° − 125° = **55°**.

In Δ DOC, ∠DCO = 180° − ∠CDO − ∠DOC = 180° − 70° − 55° = **55°**.

Δ ODC ~ Δ OBA, so corresponding angles are equal: ∠OAB = ∠OCD = **55°**.

3 **Diagonals AC and BD of a trapezium ABCD with AB || DC intersect each other at the point O. Using a similarity criterion for two triangles, show that OA/OC = OB/OD.**

✔ ANSWER

1. In Δ AOB and Δ COD: AB || DC with transversals AC and BD, so

∠OAB = ∠OCD and ∠OBA = ∠ODC (alternate angles).

2. ∠AOB = ∠COD (vertically opposite angles).

3. So Δ AOB ~ Δ COD (AA similarity).

4. Corresponding sides of similar triangles are proportional: OA/OC = OB/OD (= AB/CD).

Proved.

4 **In Fig. 6.36, QR/QS = QT/PR and ∠1 = ∠2. Show that Δ PQS ~ Δ TQR.**

✔ ANSWER

1. In Δ PQR, ∠1 = ∠2, i.e. ∠PQR = ∠PRQ, so PQ = PR (sides opposite equal angles).
2. Given QR/QS = QT/PR. Replacing PR by PQ: QR/QS = QT/PQ, i.e. QS/QR = PQ/QT.
3. In Δ PQS and Δ TQR: QS/QR = PQ/QT (from step 2) and ∠PQS = ∠TQR (same angle ∠1 = ∠Q).
4. So **Δ PQS ~ Δ TQR** by the SAS similarity criterion. Proved.

5 **S and T are points on sides PR and QR of Δ PQR such that ∠P = ∠RTS. Show that Δ RPQ ~ Δ RTS.**

✔ ANSWER

1. In Δ RPQ and Δ RTS: ∠RPQ = ∠RTS (given).
2. ∠PRQ = ∠TRS (common angle ∠R).
3. So **Δ RPQ ~ Δ RTS** by the AA similarity criterion. Proved.

6 **In Fig. 6.37, if Δ ABE ≅ Δ ACD, show that Δ ADE ~ Δ ABC.**

✔ ANSWER

1. Δ ABE ≅ Δ ACD ⇒ AB = AC and AE = AD (CPCT).

2. So AD/AB = AE/AC (both equal AD/AB, since AE = AD and AC = AB).

3. In Δ ADE and Δ ABC: AD/AB = AE/AC and the included angle ∠DAE = ∠BAC (common ∠A).

4. So **Δ ADE ~ Δ ABC** by the SAS similarity criterion. Proved.

7 **In Fig. 6.38, altitudes AD and CE of Δ ABC intersect each other at the point P. Show that: (i) Δ AEP ~ Δ CDP (ii) Δ ABD ~ Δ CBE (iii) Δ AEP ~ Δ ADB (iv) Δ PDC ~ Δ BEC**

✔ ANSWER

AD ⊥ BC and CE ⊥ AB, so ∠ADB = ∠ADC = 90° and ∠CEA = ∠CEB = 90°.

(i) In Δ AEP and Δ CDP: ∠AEP = ∠CDP = 90°; ∠APE = ∠CPD (vertically opposite). So **Δ AEP ~ Δ CDP** (AA).

(ii) In Δ ABD and Δ CBE: ∠ADB = ∠CEB = 90°; ∠ABD = ∠CBE (common ∠B). So **Δ ABD ~ Δ CBE** (AA).

(iii) In Δ AEP and Δ ADB: ∠AEP = ∠ADB = 90°; ∠EAP = ∠DAB (common ∠A). So **Δ AEP ~ Δ ADB** (AA).

(iv) In Δ PDC and Δ BEC: ∠PDC = ∠BEC = 90°; ∠PCD = ∠BCE (common ∠C). So **Δ PDC ~ Δ BEC** (AA).

8 **E is a point on the side AD produced of a parallelogram ABCD and BE intersects CD at**

F. Show that Δ ABE ~ Δ CFB.

✔ ANSWER

1. In Δ ABE and Δ CFB: ∠BAE = ∠FCB (opposite angles of parallelogram ABCD, ∠A = ∠C).

2. AE || BC (AD produced is parallel to BC) with transversal BE, so ∠AEB = ∠CBF (alternate angles).

3. So **Δ ABE ~ Δ CFB** by the AA similarity criterion. Proved.

9 **In Fig. 6.39, ABC and AMP are two right triangles, right angled at B and M respectively. Prove that: (i) Δ ABC ~ Δ AMP (ii) CA/PA = BC/MP**

✔ ANSWER

(i) In Δ ABC and Δ AMP: ∠ABC = ∠AMP = 90°; ∠BAC = ∠MAP (common ∠A).

So **Δ ABC ~ Δ AMP** by the AA similarity criterion.

(ii) Corresponding sides of similar triangles are proportional: AB/AM = BC/MP = CA/PA.

Hence **CA/PA = BC/MP**. Proved.

10 **CD and GH are respectively the bisectors of ∠ACB and ∠EGF such that D and H lie on sides AB and FE of Δ ABC and Δ EFG respectively. If Δ ABC ~ Δ FEG, show that: (i)**

CD/GH = AC/FG (ii) Δ DCB ~ Δ HGE (iii) Δ DCA ~ Δ HGF

✔ ANSWER

Δ ABC ~ Δ FEG ⇒ ∠A = ∠F, ∠B = ∠E, ∠ACB = ∠FGE.

CD and GH are bisectors, so ∠ACD = ∠DCB = ½ ∠ACB and ∠FGH = ∠HGE = ½ ∠FGE. Hence ∠ACD = ∠FGH and ∠DCB = ∠HGE.

(i) In Δ ACD and Δ FGH: ∠A = ∠F and ∠ACD = ∠FGH. So Δ ACD ~ Δ FGH (AA). Therefore **CD/GH = AC/FG**.

(ii) In Δ DCB and Δ HGE: ∠B = ∠E and ∠DCB = ∠HGE. So **Δ DCB ~ Δ HGE** (AA).

(iii) In Δ DCA and Δ HGF: ∠A = ∠F and ∠DCA = ∠HGF. So **Δ DCA ~ Δ HGF** (AA).

11 **In Fig. 6.40, E is a point on side CB produced of an isosceles triangle ABC with AB = AC. If AD ⊥ BC and EF ⊥ AC, prove that Δ ABD ~ Δ ECF.**

✔ ANSWER

1. AB = AC, so ∠ABC = ∠ACB (angles opposite equal sides), i.e. ∠ABD = ∠ECF.

2. AD ⊥ BC and EF ⊥ AC, so ∠ADB = ∠EFC = 90°.

3. In Δ ABD and Δ ECF: ∠ABD = ∠ECF and ∠ADB = ∠EFC.

4. So **Δ ABD ~ Δ ECF** by the AA similarity criterion. Proved.

12 **Sides AB and BC and median AD of a triangle ABC are respectively proportional to sides PQ and QR and median PM of Δ PQR (see Fig. 6.41). Show that Δ ABC ~ Δ PQR.**

✔ ANSWER

Given: AB/PQ = BC/QR = AD/PM. ... (1)

1. D and M are mid-points, so BC = 2BD and QR = 2QM. Then BC/QR = 2BD/2QM = BD/QM.

2. From (1): AB/PQ = BD/QM = AD/PM.

3. So Δ ABD ~ Δ PQM (SSS similarity) ⇒ ∠ABD = ∠PQM, i.e. ∠B = ∠Q.

4. In Δ ABC and Δ PQR: AB/PQ = BC/QR (given) and the included angles ∠B = ∠Q.

5. So **Δ ABC ~ Δ PQR** by the SAS similarity criterion. Proved.

13 **D is a point on the side BC of a triangle ABC such that ∠ADC = ∠BAC. Show that CA^2 = CB·CD.**

✔ ANSWER

1. In Δ ADC and Δ BAC: ∠ADC = ∠BAC (given) and ∠ACD = ∠BCA (common ∠C).

2. So Δ ADC ~ Δ BAC (AA similarity).

3. Corresponding sides are proportional: CA/CB = CD/CA.

4. Cross-multiplying, **$CA^2 = CB \cdot CD$**. Proved.

14 **Sides AB and AC and median AD of a triangle ABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Show that Δ ABC ~ Δ PQR.**

✔ ANSWER

Given: AB/PQ = AC/PR = AD/PM. ... (1)

Construction: produce AD to E so that AD = DE, and PM to N so that PM = MN. Join BE, CE, QN, RN.

1. In quadrilateral ABEC, diagonals AE and BC bisect each other (BD = DC, AD = DE), so ABEC is a parallelogram ⇒ BE = AC. Similarly PQNR is a parallelogram ⇒ QN = PR.

2. From (1): AB/PQ = BE/QN = 2AD/2PM = AE/PN. So Δ ABE ~ Δ PQN (SSS) ⇒ ∠BAE = ∠QPN. ... (2)

3. Similarly AC/PR = CE/RN = AE/PN (CE = AB, RN = PQ, sides of the parallelograms), so Δ ACE ~ Δ PRN (SSS) ⇒ ∠CAE = ∠RPN. ... (3)

4. Adding (2) and (3): ∠BAE + ∠CAE = ∠QPN + ∠RPN, i.e. ∠BAC = ∠QPR.

5. In Δ ABC and Δ PQR: AB/PQ = AC/PR (given) and the included angles ∠A = ∠P.

6. So **Δ ABC ~ Δ PQR** by the SAS similarity criterion. Proved.

15 **A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.**

✔ ANSWER

Let the height of the tower be h m. The pole, the tower and their shadows form two right triangles.

At the same time the sun makes the same angle with the ground, so the two triangles are similar (AA: 90° and the sun's angle).

Height of pole / its shadow = height of tower / its shadow:

6/4 = h/28 ⇒ h = (6 × 28)/4 = 42.

Height of the tower = **42 m**.

16 **If AD and PM are medians of triangles ABC and PQR, respectively where Δ ABC ~ Δ PQR, prove that AB/PQ = AD/PM.**

✔ ANSWER

1. Δ ABC ~ Δ PQR ⇒ AB/PQ = BC/QR and ∠B = ∠Q. ... (1)

2. AD and PM are medians, so BC = 2BD and QR = 2QM. Then BC/QR = 2BD/2QM = BD/QM.

3. From (1): AB/PQ = BD/QM, with included angles ∠ABD = ∠PQM.

4. So Δ ABD ~ Δ PQM (SAS similarity).

5. Hence **AB/PQ = AD/PM** (corresponding sides). Proved.

🎉 **Chapter 6 done! Revise the answers once more tomorrow.**